EXPERIMENT NO 8

8.1.1 Multiplication of two square matrices

Algorithm

Step 1: Start the program.

Step 2: Input the dimension n of the square matrices.

Step 3: Input the elements of the first matrix A of size n × n.

Step 4: Input the elements of the second matrix B of size n × n.

Step 5: Initialize i = 0.

Step 6: Check if i < n.

- If **No**, go to **Step 18**.
- If **Yes**, go to **Step 7**.

Step 7: Initialize j = 0.

Step 8: Check if j < n.

- If **No**, go to **Step 16**.
- If **Yes**, go to **Step 9**.

Step 9: Set C[i][j] = 0.

Step 10: Initialize k = 0.

Step 11: Check if k < n.

- If **No**, go to **Step 14**.
- If **Yes**, go to **Step 12**.

Step 12: Compute
C[i][j] = C[i][j] + A[i][k] × B[k][j].

Step 13: Increment k = k + 1 and go back to **Step 11**.

Step 14: Increment j = j + 1.

Step 15: Go back to **Step 8**.

Step 16: Increment i = i + 1.

Step 17: Go back to **Step 6**.

Step 18: Print the resultant matrix C.

Step 19: End the program.

Flowchart

EXPERIMENT NO 8

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PYTHON CODE

```
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    row = list(map(int, input().split()))
    A.append(row)
print("second matrix:")
B = []
for i in range(n):
 row = list(map(int, input().split()))
 B.append(row)
result = []
for i in range(n):
    result.append([0] * n)
for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j] += A[i][k] * B[k][j]


print("Resultant Matrix:")
for row in result:
    print(*row)
```

EXCECUTION

EXPERIMENT NO 8

CODETANTRA

8.1.1. Multiplication of Two Square Matrices

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices (n × n).
- The second line prints: "first matrix: " and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints "second matrix: " and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix: " followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices

Sample Test Cases

Multiplication

```
# Type Content here...
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    r = list(map(int, input().split()))
    A.append(r)

print("second matrix:")
B = []
for i in range(n):
    r = list(map(int, input().split()))
    B.append(r)

result = []
for i in range(n):
    result.append([0]*n)

for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j]+=A[i][k]*B[k][j]

print("resultant Matrix:")
for i in range(n):
    for j in range(n):
        if j == n - 1:
            print(result[i][j], end="")
        else:
```

Test case 1

Expected output

dimension: 3

first matrix:

1 2 3

4 5 6

7 8 9

second matrix:

1 2 3

4 5 6

7 8 9

Actual output

dimension: 3

first matrix:

1 2 3

4 5 6

7 8 9

second matrix:

1 2 3

4 5 6

7 8 9